

Camp Overview

This is a full plan for a **AI Skills for Kids** camp of **51 children** on Saturday 15 August 2026. The intake is split into age bands — 7-9 yrs (20), 10-12 yrs (19), 13-15 yrs (12) — and each band into small groups so every child works at the right level and gets plenty of hands-on time. The sections below give the group split, staffing, the master timetable, and a tailored programme for each band.

Groups at a Glance

Each group stays together all day with its own lead facilitator, so no child is ever in a group larger than the target size.

AGE BAND	GROUP	SIZE
7-9 yrs	7-9 yrs · Group A	10 children
7-9 yrs	7-9 yrs · Group B	10 children
10-12 yrs	10-12 yrs · Group A	10 children
10-12 yrs	10-12 yrs · Group B	9 children
13-15 yrs	13-15 yrs · Group A	6 children
13-15 yrs	13-15 yrs · Group B	6 children

Staffing & Ratios

With **51 children** in **6 groups**, this plan needs **10 adults** on the day — an overall ratio of about **1 adult to 5.1 children** (target 1:10 or better).

ROLE	PEOPLE
Lead facilitators (one per group)	6

ROLE	PEOPLE
Floating helpers	2
Camp coordinator	1
First-aider	1
Total adults	10

Master Day Timetable

The whole camp runs on one clock. Groups start and stop together so breaks, lunch and the showcase stay in sync.

TIME	ACTIVITY
08:30 – 09:00	Arrival, registration & name badges
09:00 – 09:20	Opening assembly — welcome, safety brief, split into groups (all bands together)
09:20 – 10:45	Project Block 1 — each group in its own space
10:45 – 11:05	Morning break & snack (staggered so the space doesn't crowd)
11:05 – 12:30	Project Block 2 — build continues / next activity
12:30 – 13:15	Lunch & free play (supervised, outdoors if possible)
13:15 – 14:30	Project Block 3 — finish builds, test & debug
14:30 – 14:50	Showcase prep — groups tidy kits and rehearse a 60-second demo
14:50 – 15:20	Showcase & demos — every group presents to the whole camp
15:20 – 15:30	Closing circle, certificates & departure

Programme by Age Band

Each band works through age-appropriate TinkerWithMe projects. Every project below includes its materials, step-by-step instructions and full code/build steps so a facilitator can run it directly.

7-9 yrs — 20 children in 2 group(s)

Projects for this band: AI art studio, Teachable Machine, Prompting power. Each group's lead facilitator preps the kit, keeps the group to the master timetable, and watches for the common mistakes noted in each project.

AI art studio (*Beginner* · ~60 min)

What We're Doing

In this workshop, students step into the role of "AI art directors" — they will use DALL-E, a text-to-image AI tool, to turn written descriptions (called prompts) into original pictures. Rather than drawing with a pencil, students will "draw with words," learning how small changes in language can dramatically change an image. By the end of the session, each student will have created a small personal gallery of AI-generated artworks and will understand, at a basic level, how AI image generators interpret text and turn it into visuals.

The activity is designed to build AI literacy through hands-on experimentation: students will notice patterns in what works and what doesn't, discover the AI's strengths and quirks, and reflect on questions of creativity, authorship, and bias in AI-generated art.

Tools & Resources

- **DALL-E (via ChatGPT with image generation):** <https://chat.openai.com> — Free accounts have a limited number of image generations per day/hour; this can reset or run out mid-lesson, so plan accordingly.
- **Bing Image Creator (powered by DALL-E):** <https://www.bing.com/images/create> — Free with a Microsoft account; uses a "boost" credit system that limits fast generations, but images can still be created (slower) after credits run out.
- **Craiyon (free, no login):** <https://www.craiyon.com> — Good backup tool if internet is slow or accounts are unavailable; lower image quality but no sign-up needed.
- **A shared document or slide deck** (Google Slides/Docs) for students to paste their prompts and screenshot their images for later review.
- **Projector or shared screen** for teacher demonstration before students work individually or in pairs.

Step-by-Step Activity

1. Open the chosen AI art tool (e.g., Bing Image Creator or ChatGPT) on the projector and explain: "This is an AI that turns your written descriptions into pictures. We call these descriptions 'prompts.'"
2. Demonstrate with a simple prompt: *"A red bicycle parked under a mango tree"*. Generate the image live and discuss what the AI got right or wrong.
3. Show how adding detail changes the result. Regenerate using: *"A red bicycle parked under a mango tree, in the style of a watercolour painting, warm afternoon light"*. Ask students to compare the two outputs.
4. Introduce the "prompt formula": **Subject + Setting + Style + Mood/ Lighting**. Write this formula somewhere visible for reference.
5. Have students log in (individually or in pairs) to their assigned tool.
6. Give students their first guided prompt to try: *"A lion wearing sunglasses, sitting on a rooftop in Nairobi, cartoon style, sunny day"*. Let them generate and view the result.
7. Ask students to write their own original prompt using the prompt formula, generate the image, and save/screenshot it.
8. Have students generate at least 2 variations of their original prompt (changing style, mood, or setting) to see how the output changes.
9. Ask each student to choose their favourite image and write one sentence explaining why the AI's interpretation surprised them or matched their expectations.
10. Bring the group back together to share 3–4 favourite images on the projector, reading the prompts aloud before revealing the image.
11. Close with a group discussion using the questions below.

Example Outputs

- **Example 1:** Prompt: *"A giraffe reading a book in a library, digital art style"* — produces a whimsical, colourful image of a giraffe with glasses holding an open book, surrounded by bookshelves. Shows good understanding of combining subject + action + setting + style.
- **Example 2:** Prompt: *"A futuristic matatu flying over Nairobi at sunset, in the style of a comic book"* — produces a vibrant, localized image blending familiar

Kenyan context with sci-fi elements. Demonstrates creative combination of local culture and imaginative "what if" thinking.

- **Example 3 (comparison set):** Same subject ("a cat drinking tea") generated in four styles — realistic photo, watercolour painting, pixel art, and pencil sketch — shown side by side. Demonstrates understanding that style keywords significantly change the output while the subject stays recognizable.

Facilitation Tips

- **Pacing:** Spend no more than 10 minutes on the live demo — students learn fastest by trying prompts themselves. Reserve at least 25–30 minutes for hands-on generation.
- **Free-tier limits:** Warn students in advance that they have a limited number of image generations. Encourage them to plan/write their prompt in a doc first, then generate, rather than generating impulsively and running out of credits.
- **Common confusion:** Students often expect the AI to read their minds. Reassure them that vague prompts produce vague or strange results — this is a teaching moment about clear communication, not an AI failure.
- **Handling odd or "wrong" outputs:** If the AI produces something inaccurate (e.g., wrong number of fingers, garbled text, mismatched objects), use it as a discussion point rather than an error to hide — ask "why do you think that happened?"
- **Sensitive content:** Remind students that these tools have content filters and will block certain prompts (violence, real people, etc.). If a prompt is blocked, ask students to rephrase rather than try to bypass the filter.
- **Grouping:** Pairing students on one device can reduce the load on free-tier limits and encourages discussion about word choice.
- **Internet issues:** Have Craiyon (no login, browser-based) ready as backup if the main tool is slow or blocked by school firewalls.

Discussion Questions

- What is a "prompt," and why does the wording of a prompt matter so much to the AI's output?
- How do you think the AI "knows" what a lion, a mango tree, or a watercolour style looks like?

- Can you think of a time today when the AI misunderstood your prompt? What might have caused that?
- Is the image the AI created "art"? Who do you think should be credited as the artist — you, the AI, or both? Why?
- AI image generators are trained on millions of real images made by human artists. What are the benefits and possible concerns with that?

Extension Challenges

- **Easy:** Generate the same subject in 3 different art styles (e.g., cartoon, realistic, watercolour) and write one sentence comparing the results.
- **Medium:** Create a short 4-image "story" using prompts that build on each other (e.g., a character's journey from morning to night), keeping the character consistent across prompts.
- **Hard:** Design a prompt for an image that represents an abstract idea (e.g., "friendship," "hope," or "climate change") without directly naming it in an obvious way, then present the image to the class and have them guess what abstract idea it represents.

Teachable Machine (*Intermediate* · ~90 min)

What We're Doing

In this hands-on activity, students will build their very own image classifier using Google's Teachable Machine — a free browser-based tool that lets anyone train a real machine learning model without writing a single line of code. Students will use their device's webcam (or uploaded photos) to teach a model to recognise different categories of images, such as hand gestures, objects, or facial expressions.

By the end of the session, students will have a working AI model that can correctly identify what it "sees" in real time, and — more importantly — they will understand the core concepts behind how machine learning actually works: collecting data, training a model, testing its accuracy, and recognising bias when the model gets things wrong. This is a foundational lesson for understanding that AI systems are only as good as the data used to train them.

This is an intermediate-level session, so it assumes students already have basic familiarity with what AI is and have perhaps seen a demo of image recognition

before. Here, they move from "watching AI work" to "building an AI model themselves."

Tools & Resources

- **Google Teachable Machine** — <https://teachablemachine.withgoogle.com/> (free, no login required, runs entirely in the browser; no data leaves the device unless exported)
- **A device with a webcam** — laptop, tablet, or desktop with a working camera. If no webcam is available, students can upload image files instead.
- **Chrome or Edge browser** (recommended for best camera permission handling; Teachable Machine may behave inconsistently on older browsers)
- **Optional: sample image folders** pre-downloaded on a USB drive or shared folder, in case internet bandwidth is limited and students need to upload images instead of using live webcam capture
- **Free-tier limits to watch for:** Teachable Machine is completely free with no account needed, but it runs entirely client-side, meaning large image classes or many samples can slow down or crash older/low-spec devices. Models are also not saved automatically to the cloud unless the student explicitly exports/downloads or uploads to Google Drive — remind students to save their work before closing the tab.

Step-by-Step Activity

1. Open a browser and navigate to teachablemachine.withgoogle.com. Click "Get Started," then choose "Image Project," and select "Standard image model."
2. Explain the interface: there are "Class" boxes on the left (where training images go), a "Training" panel in the middle, and a "Preview" panel on the right (where the live model will make predictions).
3. Rename Class 1 and Class 2 to something meaningful — for example, "Thumbs Up" and "Thumbs Down," or "Book" and "Phone." Add a third class ("Nothing" or "Background") to help the model learn what "neither category" looks like.
4. Click the webcam icon under each class box and grant camera permission when prompted by the browser.

5. For each class, hold down the "Hold to Record" button to capture at least 50–100 images while varying angle, distance, lighting, and background. Repeat for every class.
6. Click the "Train Model" button in the centre panel. Do not close the tab or switch away — training takes anywhere from 10 seconds to a few minutes depending on the number of images and device speed.
7. Once training finishes, test the model live in the "Preview" panel by holding up the object/gesture in front of the webcam and watching the confidence percentages update in real time.
8. Deliberately test with a "confusing" input — for example, an object that overlaps two classes or unusual lighting — and discuss why the model gets it wrong or right.
9. Go back and add 20–30 more training images to the class that performed worst, focusing on varied conditions, then retrain and re-test to observe improvement.
10. Export the model: click "Export Model," choose "Tensorflow.js," and either copy the shareable link or download the model files to keep a record of the finished project.
11. Have each student (or pair) present their trained model to the class for 1 minute, showing what it classifies and one thing they discovered while training it.

Example Outputs

- **Gesture Classifier:** A model trained on "Thumbs Up," "Thumbs Down," and "Open Palm" that correctly identifies each gesture with over 90% confidence, including when tested from different angles.
- **Object Sorter:** A model that distinguishes between a "Water Bottle," "Notebook," and "Phone" placed on a desk, demonstrating that the model has learned shape and colour cues rather than just background context.
- **Expression Detector:** A model trained on "Smiling Face," "Serious Face," and "Surprised Face" that reveals limitations — for instance, performing well on the student who trained it but poorly on a classmate with different facial features, sparking rich discussion on bias in training data.

Facilitation Tips

- Do a live demo first before letting students loose — show one full cycle (collect → train → test) so they see the whole workflow before diving in themselves.
- Encourage variety early: many students will collect all their training images sitting in the exact same position, which produces a model that fails the moment lighting or angle changes. Prompt them mid-activity to "move around, change the background, get closer and farther."
- Watch device performance — if a laptop is old or many browser tabs are open, training can be slow. Suggest closing other tabs before starting.
- Some students will get suspiciously perfect (100%) results initially — use this as a teaching moment: ask "would this still work if I stood by the window?" and test it live to reveal overfitting.
- Camera permission pop-ups are a common stumbling block — walk the room early to make sure everyone has granted access before the training phase begins.
- Keep training sample counts roughly balanced across classes; a common confusion is students recording 200 images for one class and 20 for another, which skews the model's predictions toward the larger class.
- Plan for roughly: 10 min setup/demo, 15 min class naming and data collection, 10 min first training and testing, 15 min data improvement and retraining, 15 min export/sharing, remaining time for presentations and discussion.
- If webcams are unavailable or unreliable, have a backup folder of pre-selected images ready so students can use the "upload" option instead of live capture.

Discussion Questions

- What steps did the model go through between you showing it images and it being able to make predictions?
- Why did the model perform differently when you changed the lighting, angle, or background — what does this tell you about how it "sees"?
- If someone else used your trained model, do you think it would work as well for them as it did for you? Why or why not?
- What is the difference between the model "memorising" your exact training images versus actually "learning" the category?

- Where in daily life might a similar image classifier already be working behind the scenes — and what could go wrong if its training data was unfair or incomplete?

Extension Challenges

- **Easy:** Add a fourth class to your existing model and retrain it — test whether adding more categories makes the model less accurate overall, and explain why.
- **Medium:** Deliberately try to "break" your model by testing it with tricky inputs (poor lighting, partial objects, mixed backgrounds). Document at least three failure cases and propose a specific fix (e.g., more images, different angles) for each one.
- **Hard:** Export your trained model's shareable link and swap with a partner. Test your partner's model using your own face/objects instead of theirs, record the accuracy drop, and write a short explanation of why the model behaves differently for a "new" user — connecting this to real-world concerns about AI systems trained on narrow or non-representative datasets.

Prompting power (*Beginner* · ~60 min)

What We're Doing

In this lesson, students discover that getting good results from an AI chatbot like ChatGPT is a skill — not luck. Many first-time users type short, vague requests and get disappointing answers, then conclude "the AI is bad." Today's activity flips that idea on its head: students will run the **same request three times**, making it more specific each time, and watch the AI's answers transform from generic to genuinely useful.

By the end of the session, students will have created a personal "Prompt Recipe Card" — a simple reusable formula for writing clear prompts — and used it to solve a real request of their own choosing (e.g. a birthday party idea, a study plan, a story). The core takeaway: *the quality of your question shapes the quality of the AI's answer.*

Tools & Resources

- **ChatGPT (free tier)** — <https://chat.openai.com>. Requires an account (email or Google sign-in). Free tier uses GPT-3.5/GPT-4o-mini depending on availability;

may show "busy" messages at peak times — have students refresh or wait 30 seconds.

- Alternative if ChatGPT access is limited: **Microsoft Copilot** — <https://copilot.microsoft.com> (free, no login required for basic chat on some regions) or **Google Gemini** — <https://gemini.google.com>.
- Printed or digital **Prompt Recipe Card template** (teacher to prepare — a simple table with columns: Task / Role / Detail / Format).
- Shared class document (Google Doc or physical flipchart) to collect students' "before and after" prompts for comparison.
- **Free-tier watch-outs:** message limits during high traffic, occasional login/verification prompts, and slower response times when many devices are on the same Wi-Fi network — test connectivity beforehand.

Step-by-Step Activity

1. Open with a quick poll: ask students to guess what happens if you ask ChatGPT "Tell me about dogs" versus a very specific question. Write both guesses on the board.
2. Have every student open ChatGPT (or the alternative tool) and type this exact vague prompt: `Tell me about dogs`. Let them read the response silently for 1 minute.
3. As a class, discuss: Is this answer useful? What's missing? Who is it "for"? (It's generic — not written for anyone in particular.)
4. Introduce the **4-part Prompt Recipe**: *Role* (who should the AI act as?), *Task* (what exactly do you want?), *Detail* (context, audience, constraints), *Format* (list, story, table, short paragraph).
5. Model rewriting the dog prompt live using the recipe: `You are a vet. Explain to a curious 10-year-old why dogs wag their tails, in 3 fun facts with emojis.` Run it and compare the output to the first attempt.
6. Students rewrite their own version of the dog prompt using the recipe and run it. Ask 2–3 volunteers to read their prompt and result aloud.
7. Now students choose their own topic (a hobby, a school subject, a problem they have) and write a **Level 1 prompt** (vague, one sentence) and run it.
8. Students then write a **Level 2 prompt** on the same topic using the 4-part recipe, and run it.

9. Students write a **Level 3 prompt** adding one more constraint (word limit, tone, or example format) to push the answer further, and run it.
10. Students fill in their Prompt Recipe Card with their best Level 3 prompt and paste/write their favourite AI response next to it.
11. Close with a gallery walk or quick share-out: students show their "before" (Level 1) and "after" (Level 3) prompts to a partner and explain what changed.

Example Outputs

- **Level 1 prompt:** "Help me plan a party." → Generic bullet list of party ideas with no context.
Level 3 prompt: "You are an event planner. Plan a budget-friendly 12th birthday party for 15 kids in Nairobi, with an outdoor theme, under KES 5,000. Give me a table with activity, time, and cost." → A structured table with realistic local activities, timings, and costs.
- **Level 1 prompt:** "Tell me about the water cycle." → A dense, textbook-style paragraph.
Level 3 prompt: "You are a fun science teacher. Explain the water cycle to a 9-year-old using a short story about a raindrop named Dudu, in 5 sentences." → An engaging, age-appropriate mini-story that's easy to remember.
- **Level 1 prompt:** "Give me a study plan." → Vague, one-size-fits-all list.
Level 3 prompt: "You are a study coach. Create a 5-day revision timetable for a Grade 7 student preparing for a Maths exam on fractions and decimals, with 30-minute sessions and one fun break each day. Present it as a table." → A clear, personalized, actionable schedule.

Facilitation Tips

- **Pacing:** Keep the first vague-prompt exercise short (under 5 minutes) — its purpose is just to create the "aha, that's not very useful" moment, not to analyze deeply.
- **Common confusion:** Students often think longer prompts automatically mean better answers. Clarify that *specificity*, not length, is the key — a short but precise prompt beats a long rambling one.
- Some students will get an unexpectedly poor or off-topic response even with a good prompt. Use this as a teaching moment: "AI doesn't always get it right first

time — what could we add or change to fix this?" Encourage iterating rather than giving up.

- Watch for students copying a classmate's exact prompt instead of writing their own — walk around and prompt them to personalize the topic.
- If the free tool is slow or down, pair students up to share one working device rather than stalling the whole class.
- Remind students that any personal information typed into the chat isn't private — avoid entering real names, addresses, or school details in prompts.
- For younger or less confident writers, provide the Recipe Card as fill-in-the-blank sentence starters rather than a blank table.

Discussion Questions

- What is the difference between a "vague" prompt and a "specific" prompt? Can you give an example of each?
- Why do you think giving the AI a "role" (like "you are a vet" or "you are a teacher") changes its answer?
- Was there ever a moment when a more detailed prompt still gave you a wrong or unhelpful answer? What did you do next?
- Do you think there's such a thing as a prompt that's "too detailed" or "too long"? Why or why not?
- How is asking a good question to an AI similar to or different from asking a good question to a teacher or friend?

Extension Challenges

- **Easy:** Take one of your Level 3 prompts and change only the *format* instruction (e.g. ask for a poem instead of a list). Compare how differently the same information is presented.
- **Medium:** Write a prompt that asks the AI to give you its answer in two different styles at once (e.g. "Explain this first like I'm 5 years old, then like I'm 15 years old") and compare the two explanations.
- **Hard:** Design a "prompt chain" — a sequence of 3 connected prompts where each one builds on the AI's previous answer (e.g. first ask for a story idea, then ask it to turn that idea into a script, then ask it to suggest a title and moral). Present your final result and explain how each prompt shaped the next.

10-12 yrs — 19 children in 2 group(s)

Projects for this band: Teachable Machine, AI art studio, AI storytelling. Each group's lead facilitator preps the kit, keeps the group to the master timetable, and watches for the common mistakes noted in each project.

Teachable Machine (*Intermediate* · ~90 min)

What We're Doing

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The activity is designed to build AI literacy through hands-on experimentation: students will notice patterns in what works and what doesn't, discover the AI's strengths and quirks, and reflect on questions of creativity, authorship, and bias in AI-generated art.

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- **Craiyon (free, no login):** <https://www.craiyon.com> — Good backup tool if internet is slow or accounts are unavailable; lower image quality but no sign-up needed.
- **A shared document or slide deck** (Google Slides/Docs) for students to paste their prompts and screenshot their images for later review.
- **Projector or shared screen** for teacher demonstration before students work individually or in pairs.

Step-by-Step Activity

1. Open the chosen AI art tool (e.g., Bing Image Creator or ChatGPT) on the projector and explain: "This is an AI that turns your written descriptions into pictures. We call these descriptions 'prompts.'"
2. Demonstrate with a simple prompt: *"A red bicycle parked under a mango tree"*. Generate the image live and discuss what the AI got right or wrong.
3. Show how adding detail changes the result. Regenerate using: *"A red bicycle parked under a mango tree, in the style of a watercolour painting, warm afternoon light"*. Ask students to compare the two outputs.
4. Introduce the "prompt formula": **Subject + Setting + Style + Mood/ Lighting**. Write this formula somewhere visible for reference.
5. Have students log in (individually or in pairs) to their assigned tool.
6. Give students their first guided prompt to try: *"A lion wearing sunglasses, sitting on a rooftop in Nairobi, cartoon style, sunny day"*. Let them generate and view the result.
7. Ask students to write their own original prompt using the prompt formula, generate the image, and save/screenshot it.
8. Have students generate at least 2 variations of their original prompt (changing style, mood, or setting) to see how the output changes.
9. Ask each student to choose their favourite image and write one sentence explaining why the AI's interpretation surprised them or matched their expectations.
10. Bring the group back together to share 3–4 favourite images on the projector, reading the prompts aloud before revealing the image.
11. Close with a group discussion using the questions below.

Example Outputs

- **Example 1:** Prompt: *"A giraffe reading a book in a library, digital art style"* — produces a whimsical, colourful image of a giraffe with glasses holding an open book, surrounded by bookshelves. Shows good understanding of combining subject + action + setting + style.
- **Example 2:** Prompt: *"A futuristic matatu flying over Nairobi at sunset, in the style of a comic book"* — produces a vibrant, localized image blending familiar

Kenyan context with sci-fi elements. Demonstrates creative combination of local culture and imaginative "what if" thinking.

- **Example 3 (comparison set):** Same subject ("a cat drinking tea") generated in four styles — realistic photo, watercolour painting, pixel art, and pencil sketch — shown side by side. Demonstrates understanding that style keywords significantly change the output while the subject stays recognizable.

Facilitation Tips

- **Pacing:** Spend no more than 10 minutes on the live demo — students learn fastest by trying prompts themselves. Reserve at least 25–30 minutes for hands-on generation.
- **Free-tier limits:** Warn students in advance that they have a limited number of image generations. Encourage them to plan/write their prompt in a doc first, then generate, rather than generating impulsively and running out of credits.
- **Common confusion:** Students often expect the AI to read their minds. Reassure them that vague prompts produce vague or strange results — this is a teaching moment about clear communication, not an AI failure.
- **Handling odd or "wrong" outputs:** If the AI produces something inaccurate (e.g., wrong number of fingers, garbled text, mismatched objects), use it as a discussion point rather than an error to hide — ask "why do you think that happened?"
- **Sensitive content:** Remind students that these tools have content filters and will block certain prompts (violence, real people, etc.). If a prompt is blocked, ask students to rephrase rather than try to bypass the filter.
- **Grouping:** Pairing students on one device can reduce the load on free-tier limits and encourages discussion about word choice.
- **Internet issues:** Have Craiyon (no login, browser-based) ready as backup if the main tool is slow or blocked by school firewalls.

Discussion Questions

- What is a "prompt," and why does the wording of a prompt matter so much to the AI's output?
- How do you think the AI "knows" what a lion, a mango tree, or a watercolour style looks like?

- Can you think of a time today when the AI misunderstood your prompt? What might have caused that?
- Is the image the AI created "art"? Who do you think should be credited as the artist — you, the AI, or both? Why?
- AI image generators are trained on millions of real images made by human artists. What are the benefits and possible concerns with that?

Extension Challenges

- **Easy:** Generate the same subject in 3 different art styles (e.g., cartoon, realistic, watercolour) and write one sentence comparing the results.
- **Medium:** Create a short 4-image "story" using prompts that build on each other (e.g., a character's journey from morning to night), keeping the character consistent across prompts.
- **Hard:** Design a prompt for an image that represents an abstract idea (e.g., "friendship," "hope," or "climate change") without directly naming it in an obvious way, then present the image to the class and have them guess what abstract idea it represents.

AI storytelling (*Beginner* · ~75 min)

What We're Doing

In this lesson, students become author-illustrator teams working alongside AI. Each student (or pair) will invent a short story — a folktale, adventure, or funny anecdote — and use a text-based AI tool to help draft and polish the writing. They will then use an AI image generator to create illustrations that match their story's characters, settings, and mood. By the end of the session, each student will have a complete illustrated mini-story (3–6 pages) that they can present to the class, either as a simple printed booklet, a slideshow, or a digital document.

The goal is not just to produce a cute story, but to help students understand how AI can act as a creative collaborator — generating ideas, suggesting words, and turning descriptions into pictures — while the human remains the director, editor, and decision-maker. Students will practise clear instruction-writing (prompting), critical evaluation of AI output, and basic story structure (beginning, middle, end).

Tools & Resources

- **ChatGPT (free tier)** — <https://chat.openai.com> — used for brainstorming plot ideas, drafting story text, and rewriting sentences. Free tier may show "usage limit reached" messages during busy periods; have a backup device or pair students up if this happens.
- **Microsoft Designer / Bing Image Creator** — <https://designer.microsoft.com> — used to generate illustrations from text descriptions. Free accounts get a limited number of "boosted" fast generations per day (typically 15–25); after that, images generate more slowly. Requires a free Microsoft account and sign-in.
- **Canva (free tier)** — <https://canva.com> — optional, for assembling the story text and AI images into a neat illustrated booklet or slideshow layout.
- **Google Docs or physical notebook** — for drafting and organising story text before pasting into design tools.
- Printed story-planning worksheet (one per student/pair) with boxes for: Character, Setting, Problem, Ending.

Connectivity note: Both AI tools require stable internet. If bandwidth is limited, consider having only the teacher or one device per group generate images, projecting results for shared discussion, while individual students focus on writing.

Step-by-Step Activity

1. **Warm-up discussion (5 min):** Ask "What makes a story fun to read or a picture fun to look at?" List student answers on the board (characters, surprises, colour, humour).
2. **Story planning (10 min):** Hand out the planning worksheet. Each student/pair fills in: Main Character (name + description), Setting (where/when), Problem (what goes wrong), and Ending (how it's solved). Example: Character — "Amina, a clever tortoise"; Setting — "a dry savanna during a drought"; Problem — "the animals can't find water"; Ending — "Amina remembers a hidden spring."
3. **Draft the story with AI (15 min):** Students open ChatGPT and enter a prompt combining their planning notes, e.g.: *"Write a short 5-paragraph children's story about a clever tortoise named Amina who helps thirsty animals find water during a drought in the savanna. Use simple language suitable for a young reader. End with a lesson about teamwork."* Encourage students to read the

output and ask AI to change things: *"Make it funnier,"* or *"Add a mischievous hyena character."*

4. **Edit and personalise (10 min):** Students copy the AI draft into a document and edit it in their own words — changing at least 3 sentences, adding a personal detail, or fixing anything that doesn't match their original plan. Emphasise: the AI text is a first draft, not the final answer.
5. **Break story into scenes (5 min):** Students split their story into 3–5 short scenes/pages, one or two sentences each, to prepare for illustration.
6. **Generate illustrations (20 min):** For each scene, students write an image prompt describing character, setting, action, and style, e.g.: *"A friendly cartoon tortoise with a red bandana walking across a dry, cracked savanna under a hot sun, children's storybook illustration style, bright colours."* Generate the image in Microsoft Designer/Bing Image Creator, review results, and regenerate with adjusted prompts if needed (e.g., "make the tortoise smaller," "add other animals in the background").
7. **Assemble the illustrated story (10 min):** Paste text and matching images into Canva (or arrange printed pages) to create the final illustrated booklet or slideshow, in correct scene order.
8. **Share out (5–10 min):** Volunteers read their story aloud or show their slides to the class.

Example Outputs

- **"Amina the Clever Tortoise":** A 4-page story about a tortoise who solves a drought crisis using memory and teamwork, illustrated with warm savanna-toned cartoon images showing the tortoise, a worried elephant, and a hidden spring.
- **"The Boy Who Talked to Matatus":** A humorous 5-scene story set in Nairobi about a boy who discovers his neighbourhood matatu can talk and give him life advice, illustrated in a bright, comic-style with colourful matatu artwork and city backgrounds.
- **"Nia and the Missing Stars":** A short fantasy story where a girl uses a magic telescope to return stolen stars to the night sky, illustrated with a dreamy blue-and-purple palette, showing character growth from worried to confident across three scenes.

Facilitation Tips

- Model the full process once on a projector before students start, showing both a text prompt and an image prompt, including what to do when the AI gives an unexpected or "wrong" result (simply rewrite the prompt — this is normal and expected, not a failure).
- Keep an eye on pacing: writing tends to take longer than planned, illustration less so. Use a visible timer for each stage to keep the group moving together.
- If AI-generated character appearance changes between images (e.g., the tortoise looks different in each scene), turn this into a teaching moment about AI's lack of memory between separate image requests — discuss how professional users solve this (character reference prompts, consistent descriptive keywords).
- Some students may want to copy the entire AI story unedited — gently require the "3 sentence change" rule so every submission has a personal authorial touch.
- Watch for inappropriate or odd image results (common with vague prompts); encourage specific, respectful language and review the image gallery on a shared screen if needed before students save their favourite image.
- Pair weaker readers/writers with confident peers so both contribute to writing and prompting.

Discussion Questions

- What did the AI do well when writing your story, and what did you have to change or fix yourself?
- How is asking an AI for a picture different from asking a human artist to draw one for you?
- Why did the AI sometimes draw your character differently in each image, even though you used the same name?
- Who is the "author" of your final story — you, the AI, or both? Explain your answer.
- If you used a vague prompt versus a very detailed one, how did the results differ? What does this teach us about giving instructions to AI?

Extension Challenges

- **Easy:** Add a second illustrated scene to your story that wasn't in your original plan, and write one new sentence of dialogue for your main character.
- **Medium:** Rewrite your story in a different genre (turn your adventure into a mystery, or your fairy tale into science fiction) using AI to help brainstorm the changes, then generate new matching illustrations.
- **Hard:** Create a "choose your own adventure" version of your story with at least two branching points, where the reader picks what the character does next — requiring multiple AI-generated text branches and matching illustrations for each path.

13-15 yrs — 12 children in 2 group(s)

Projects for this band: Teachable Machine, AI art studio, Bias detective. Each group's lead facilitator preps the kit, keeps the group to the master timetable, and watches for the common mistakes noted in each project.

Teachable Machine (*Intermediate* · ~90 min)

What We're Doing

In this hands-on activity, students will build their very own image classifier using Google's Teachable Machine — a free browser-based tool that lets anyone train a real machine learning model without writing a single line of code. Students will use their device's webcam (or uploaded photos) to teach a model to recognise different categories of images, such as hand gestures, objects, or facial expressions.

By the end of the session, students will have a working AI model that can correctly identify what it "sees" in real time, and — more importantly — they will understand the core concepts behind how machine learning actually works: collecting data, training a model, testing its accuracy, and recognising bias when the model gets things wrong. This is a foundational lesson for understanding that AI systems are only as good as the data used to train them.

This is an intermediate-level session, so it assumes students already have basic familiarity with what AI is and have perhaps seen a demo of image recognition before. Here, they move from "watching AI work" to "building an AI model themselves."

Tools & Resources

- **Google Teachable Machine** — <https://teachablemachine.withgoogle.com/> (free, no login required, runs entirely in the browser; no data leaves the device unless exported)
- **A device with a webcam** — laptop, tablet, or desktop with a working camera. If no webcam is available, students can upload image files instead.
- **Chrome or Edge browser** (recommended for best camera permission handling; Teachable Machine may behave inconsistently on older browsers)
- **Optional: sample image folders** pre-downloaded on a USB drive or shared folder, in case internet bandwidth is limited and students need to upload images instead of using live webcam capture
- **Free-tier limits to watch for:** Teachable Machine is completely free with no account needed, but it runs entirely client-side, meaning large image classes or many samples can slow down or crash older/low-spec devices. Models are also not saved automatically to the cloud unless the student explicitly exports/downloads or uploads to Google Drive — remind students to save their work before closing the tab.

Step-by-Step Activity

1. Open a browser and navigate to teachablemachine.withgoogle.com. Click "Get Started," then choose "Image Project," and select "Standard image model."
2. Explain the interface: there are "Class" boxes on the left (where training images go), a "Training" panel in the middle, and a "Preview" panel on the right (where the live model will make predictions).
3. Rename Class 1 and Class 2 to something meaningful — for example, "Thumbs Up" and "Thumbs Down," or "Book" and "Phone." Add a third class ("Nothing" or "Background") to help the model learn what "neither category" looks like.
4. Click the webcam icon under each class box and grant camera permission when prompted by the browser.
5. For each class, hold down the "Hold to Record" button to capture at least 50–100 images while varying angle, distance, lighting, and background. Repeat for every class.

6. Click the "Train Model" button in the centre panel. Do not close the tab or switch away — training takes anywhere from 10 seconds to a few minutes depending on the number of images and device speed.
7. Once training finishes, test the model live in the "Preview" panel by holding up the object/gesture in front of the webcam and watching the confidence percentages update in real time.
8. Deliberately test with a "confusing" input — for example, an object that overlaps two classes or unusual lighting — and discuss why the model gets it wrong or right.
9. Go back and add 20–30 more training images to the class that performed worst, focusing on varied conditions, then retrain and re-test to observe improvement.
10. Export the model: click "Export Model," choose "Tensorflow.js," and either copy the shareable link or download the model files to keep a record of the finished project.
11. Have each student (or pair) present their trained model to the class for 1 minute, showing what it classifies and one thing they discovered while training it.

Example Outputs

- **Gesture Classifier:** A model trained on "Thumbs Up," "Thumbs Down," and "Open Palm" that correctly identifies each gesture with over 90% confidence, including when tested from different angles.
- **Object Sorter:** A model that distinguishes between a "Water Bottle," "Notebook," and "Phone" placed on a desk, demonstrating that the model has learned shape and colour cues rather than just background context.
- **Expression Detector:** A model trained on "Smiling Face," "Serious Face," and "Surprised Face" that reveals limitations — for instance, performing well on the student who trained it but poorly on a classmate with different facial features, sparking rich discussion on bias in training data.

Facilitation Tips

- Do a live demo first before letting students loose — show one full cycle (collect → train → test) so they see the whole workflow before diving in themselves.

- Encourage variety early: many students will collect all their training images sitting in the exact same position, which produces a model that fails the moment lighting or angle changes. Prompt them mid-activity to "move around, change the background, get closer and farther."
- Watch device performance — if a laptop is old or many browser tabs are open, training can be slow. Suggest closing other tabs before starting.
- Some students will get suspiciously perfect (100%) results initially — use this as a teaching moment: ask "would this still work if I stood by the window?" and test it live to reveal overfitting.
- Camera permission pop-ups are a common stumbling block — walk the room early to make sure everyone has granted access before the training phase begins.
- Keep training sample counts roughly balanced across classes; a common confusion is students recording 200 images for one class and 20 for another, which skews the model's predictions toward the larger class.
- Plan for roughly: 10 min setup/demo, 15 min class naming and data collection, 10 min first training and testing, 15 min data improvement and retraining, 15 min export/sharing, remaining time for presentations and discussion.
- If webcams are unavailable or unreliable, have a backup folder of pre-selected images ready so students can use the "upload" option instead of live capture.

Discussion Questions

- What steps did the model go through between you showing it images and it being able to make predictions?
- Why did the model perform differently when you changed the lighting, angle, or background — what does this tell you about how it "sees"?
- If someone else used your trained model, do you think it would work as well for them as it did for you? Why or why not?
- What is the difference between the model "memorising" your exact training images versus actually "learning" the category?
- Where in daily life might a similar image classifier already be working behind the scenes — and what could go wrong if its training data was unfair or incomplete?

Extension Challenges

- **Easy:** Add a fourth class to your existing model and retrain it — test whether adding more categories makes the model less accurate overall, and explain why.
- **Medium:** Deliberately try to "break" your model by testing it with tricky inputs (poor lighting, partial objects, mixed backgrounds). Document at least three failure cases and propose a specific fix (e.g., more images, different angles) for each one.
- **Hard:** Export your trained model's shareable link and swap with a partner. Test your partner's model using your own face/objects instead of theirs, record the accuracy drop, and write a short explanation of why the model behaves differently for a "new" user — connecting this to real-world concerns about AI systems trained on narrow or non-representative datasets.

AI art studio (*Beginner* · ~60 min)

What We're Doing

In this workshop, students step into the role of "AI art directors" — they will use DALL-E, a text-to-image AI tool, to turn written descriptions (called prompts) into original pictures. Rather than drawing with a pencil, students will "draw with words," learning how small changes in language can dramatically change an image. By the end of the session, each student will have created a small personal gallery of AI-generated artworks and will understand, at a basic level, how AI image generators interpret text and turn it into visuals.

The activity is designed to build AI literacy through hands-on experimentation: students will notice patterns in what works and what doesn't, discover the AI's strengths and quirks, and reflect on questions of creativity, authorship, and bias in AI-generated art.

Tools & Resources

- **DALL-E (via ChatGPT with image generation):** <https://chat.openai.com> — Free accounts have a limited number of image generations per day/hour; this can reset or run out mid-lesson, so plan accordingly.

- **Bing Image Creator (powered by DALL-E):** <https://www.bing.com/images/create> — Free with a Microsoft account; uses a "boost" credit system that limits fast generations, but images can still be created (slower) after credits run out.
- **Craiyon (free, no login):** <https://www.craiyon.com> — Good backup tool if internet is slow or accounts are unavailable; lower image quality but no sign-up needed.
- **A shared document or slide deck** (Google Slides/Docs) for students to paste their prompts and screenshot their images for later review.
- **Projector or shared screen** for teacher demonstration before students work individually or in pairs.

Step-by-Step Activity

1. Open the chosen AI art tool (e.g., Bing Image Creator or ChatGPT) on the projector and explain: "This is an AI that turns your written descriptions into pictures. We call these descriptions 'prompts.'"
2. Demonstrate with a simple prompt: *"A red bicycle parked under a mango tree"*. Generate the image live and discuss what the AI got right or wrong.
3. Show how adding detail changes the result. Regenerate using: *"A red bicycle parked under a mango tree, in the style of a watercolour painting, warm afternoon light"*. Ask students to compare the two outputs.
4. Introduce the "prompt formula": **Subject + Setting + Style + Mood/ Lighting**. Write this formula somewhere visible for reference.
5. Have students log in (individually or in pairs) to their assigned tool.
6. Give students their first guided prompt to try: *"A lion wearing sunglasses, sitting on a rooftop in Nairobi, cartoon style, sunny day"*. Let them generate and view the result.
7. Ask students to write their own original prompt using the prompt formula, generate the image, and save/screenshot it.
8. Have students generate at least 2 variations of their original prompt (changing style, mood, or setting) to see how the output changes.
9. Ask each student to choose their favourite image and write one sentence explaining why the AI's interpretation surprised them or matched their expectations.

10. Bring the group back together to share 3–4 favourite images on the projector, reading the prompts aloud before revealing the image.
11. Close with a group discussion using the questions below.

Example Outputs

- **Example 1:** Prompt: *"A giraffe reading a book in a library, digital art style"* — produces a whimsical, colourful image of a giraffe with glasses holding an open book, surrounded by bookshelves. Shows good understanding of combining subject + action + setting + style.
- **Example 2:** Prompt: *"A futuristic matatu flying over Nairobi at sunset, in the style of a comic book"* — produces a vibrant, localized image blending familiar Kenyan context with sci-fi elements. Demonstrates creative combination of local culture and imaginative "what if" thinking.
- **Example 3 (comparison set):** Same subject ("a cat drinking tea") generated in four styles — realistic photo, watercolour painting, pixel art, and pencil sketch — shown side by side. Demonstrates understanding that style keywords significantly change the output while the subject stays recognizable.

Facilitation Tips

- **Pacing:** Spend no more than 10 minutes on the live demo — students learn fastest by trying prompts themselves. Reserve at least 25–30 minutes for hands-on generation.
- **Free-tier limits:** Warn students in advance that they have a limited number of image generations. Encourage them to plan/write their prompt in a doc first, then generate, rather than generating impulsively and running out of credits.
- **Common confusion:** Students often expect the AI to read their minds. Reassure them that vague prompts produce vague or strange results — this is a teaching moment about clear communication, not an AI failure.
- **Handling odd or "wrong" outputs:** If the AI produces something inaccurate (e.g., wrong number of fingers, garbled text, mismatched objects), use it as a discussion point rather than an error to hide — ask "why do you think that happened?"
- **Sensitive content:** Remind students that these tools have content filters and will block certain prompts (violence, real people, etc.). If a prompt is blocked, ask students to rephrase rather than try to bypass the filter.

- **Grouping:** Pairing students on one device can reduce the load on free-tier limits and encourages discussion about word choice.
- **Internet issues:** Have Craiyon (no login, browser-based) ready as backup if the main tool is slow or blocked by school firewalls.

Discussion Questions

- What is a "prompt," and why does the wording of a prompt matter so much to the AI's output?
- How do you think the AI "knows" what a lion, a mango tree, or a watercolour style looks like?
- Can you think of a time today when the AI misunderstood your prompt? What might have caused that?
- Is the image the AI created "art"? Who do you think should be credited as the artist — you, the AI, or both? Why?
- AI image generators are trained on millions of real images made by human artists. What are the benefits and possible concerns with that?

Extension Challenges

- **Easy:** Generate the same subject in 3 different art styles (e.g., cartoon, realistic, watercolour) and write one sentence comparing the results.
- **Medium:** Create a short 4-image "story" using prompts that build on each other (e.g., a character's journey from morning to night), keeping the character consistent across prompts.
- **Hard:** Design a prompt for an image that represents an abstract idea (e.g., "friendship," "hope," or "climate change") without directly naming it in an obvious way, then present the image to the class and have them guess what abstract idea it represents.

Bias detective (*Intermediate* · ~75 min)

What We're Doing

In this activity, students become "bias detectives" who investigate how AI tools can produce unfair, stereotyped, or unbalanced results. Bias in AI happens when a tool's training data over-represents some groups, cultures, or viewpoints and

under-represents others — and this often shows up in surprising ways in images, text, and translations. Students will run a series of carefully designed prompts through AI image generators, chatbots, and translation tools, record what they notice, and build an evidence-based "Bias Report" that identifies patterns, guesses at root causes, and proposes what a fairer tool might look like. The goal is not to make students distrust AI, but to help them use it more critically — recognising that AI reflects the data and choices of the humans who built it, and that "what the AI shows me" is not the same as "what is true or normal."

By the end of the session, each student or team will have a completed Bias Report containing at least 4 pieces of evidence (screenshots or copied text), a short written analysis, and one concrete improvement suggestion.

Tools & Resources

- **AI Image Generator:** Bing Image Creator (<https://www.bing.com/images/create>) — free with a Microsoft account; has a daily "boost" credit limit, after which images generate more slowly. Alternative: Craiyon (<https://www.craiyon.com/>) — free, no login, but slower and lower image quality.
- **Chatbot:** ChatGPT free tier (<https://chat.openai.com/>) — requires an account; free tier may have usage caps during busy periods. Alternative: Microsoft Copilot (<https://copilot.microsoft.com/>), free with Microsoft account.
- **Translation Tool:** Google Translate (<https://translate.google.com/>) — free, no login needed.
- **Recording template:** A simple shared document or printed worksheet (Google Docs/Slides or paper) titled "Bias Detective Report" with columns for Tool Used, Prompt, What We Expected, What We Got, Possible Bias Spotted.
- **Optional:** Screenshot tool (built into device) to capture evidence for the report.

Connectivity note: If internet is unreliable, pre-download 8–10 example AI outputs (images and text) from a prior session to use as backup evidence for discussion.

Step-by-Step Activity

1. Introduce the term "bias" using an everyday example (e.g., "If you only ever saw photos of doctors who were men, you might start to think all doctors are men —

- even though that's not true. AI can learn the same kind of mistaken pattern from the data it's trained on.")
2. Explain the mission: teams will test AI tools with the same prompt run multiple times or with slightly changed wording, and hunt for patterns that reveal bias.
 3. Split students into teams of 2–3. Assign each team a Bias Detective Report template to fill in as they go.
 4. Round 1 — Image Bias: Have each team enter this prompt into the image generator: *"Generate an image of a leader."* Then repeat with: *"Generate an image of a nurse," "Generate an image of a family,"* and *"Generate an image of a beautiful home."* Ask teams to record: Who/what appears? What is missing? Would this look different if drawn by someone from Nairobi?
 5. Round 2 — Text Bias: Have teams enter into the chatbot: *"Write a short story about a successful entrepreneur"* and *"Write a short story about someone who lives in a village."* Ask teams to note the names, genders, settings, and tone the AI chooses by default, and whether it makes assumptions.
 6. Round 3 — Translation Bias: Have teams enter gender-neutral sentences (in English or Swahili where the pronoun is not specified, e.g., "The doctor said the nurse should call the engineer") into Google Translate to and from a language with gendered pronouns (e.g., English ↔ French, or Swahili "yeye" ↔ English "he/she"). Ask teams to note which gender the tool assigns to each profession by default.
 7. Regroup as a class. Each team shares one striking piece of evidence from their report (2 minutes per team max).
 8. Facilitate a discussion (see Discussion Questions) on why these patterns might occur and who they might affect.
 9. Each team writes a final "Analysis" paragraph in their report: What pattern did you find? Who might this affect and how? What might have caused it?
 10. Each team writes one "Fix It" suggestion: one concrete idea for how the tool or its training data could be changed to reduce this bias (e.g., "the AI should ask for more details before assuming a gender/name" or "the image tool should show a wider range of homes and leaders").
 11. Optional: Teams present their full Bias Report to the class in a 3-minute lightning talk.

Example Outputs

Example 1 (Image Bias): A team prompts "Generate an image of a leader" three times and finds all three images show men in suits, mostly with lighter skin tones, in Western-style office settings. Their report notes: "We expected maybe some variety, like a woman chief or a local community leader in African dress, but we got the same type of image each time. This could make people think leaders always look one way."

Example 2 (Text Bias): A team asks the chatbot for a story about an entrepreneur and gets a story about "James, who built a tech startup in Silicon Valley." They then ask for a story about a village and get a story focused on poverty and hardship rather than innovation or community strength. Their analysis: "The AI seems to link success with cities and Western names, and villages only with struggle — that's not the full picture of village life."

Example 3 (Translation Bias): A team translates a gender-neutral sentence about "the doctor" and "the nurse" and finds the tool automatically assigns "he" to the doctor and "she" to the nurse even though no gender was stated. Their Fix It suggestion: "The tool could ask 'do you know the gender of this person?' instead of guessing based on the job."

Facilitation Tips

- Run one round together as a whole class first (the "leader" image prompt works well) so all students see how to record evidence before working independently.
- Keep prompts identical across teams for Round 1 so results can be compared side-by-side during the class discussion — this makes patterns much more visible and convincing.
- Watch the clock: cap each round at 10–12 minutes including recording time, or teams will run out of time for the analysis and discussion, which is the most valuable part.
- Some students may get a surprisingly diverse or "correct" result on their first try — this is a good teaching moment. Have them try 2–3 more times with the same prompt to see if the tool is truly consistent or if it varies by chance.
- Be ready to reframe unexpected results: if an AI tool produces something biased in an uncomfortable way (e.g., stereotypical or offensive content), pause and

use it as a direct teaching example rather than skipping past it — frame it calmly as "this is exactly the kind of thing we're looking for."

- Encourage students to compare AI outputs with real Nairobi/Kenyan life — this personal connection makes bias much easier to spot and discuss meaningfully.
- If a tool is slow or hits a free-tier limit, have backup screenshots ready so the lesson doesn't stall.

Discussion Questions

1. What is "bias," and where do you think an AI tool "learns" it from?
2. Which piece of evidence from today surprised you the most, and why?
3. Can an AI tool be biased even if no human intended it to be? How?
4. Who might be harmed if people believed AI-generated images or stories represented "normal" or "typical" people and places?
5. If you were building an AI tool, what is one thing you would do differently to make it fairer?

Extension Challenges

- **Easy:** Test one more prompt category not covered in class (e.g., "Generate an image of a scientist" or "Write a story about a rich person") and add it to your Bias Report with the same Tool/Prompt/Expected/Got/Bias columns.
- **Medium:** Rewrite one of your original prompts to try to reduce the bias you found (e.g., add "from different countries and backgrounds" or "with a range of ages and genders") and compare the new results to the old ones. Write a short paragraph on whether the rewritten prompt worked.
- **Hard:** Design a short "Fairness Test" — a set of 3 prompts that any AI image or text tool could be run through to check for a specific type of bias (e.g., gender-in-professions, urban-vs-rural, or skin tone diversity). Write clear instructions for how someone else could use your test and how to score the results as "low," "medium," or "high" bias.

Materials — Scaling

Each project's materials list above is **per group**. Multiply it by the number of groups in that band to get the totals to prepare. Shared kit (laptops, power strips, extension leads) can rotate between groups rather than be bought per group.

Safety, Logistics & Contingencies

- **Space:** one table per group with room to move; keep walkways clear and power leads taped down.
- **Movement:** groups start and stop together on the master timetable — stagger the break and lunch queues so the space never crowds.
- **First aid & allergies:** collect allergy info at registration; the first-aider holds the kit and the contact list all day.
- **Wet weather:** move outdoor free-play to the hall; keep one indoor back-up game ready.
- **Early finishers:** use each project's extension challenges (listed in the project pages below).
- **Equipment failure:** keep one spare kit per band; a failed build becomes a debugging lesson, not a lost hour.

Staff Briefing Checklist

- Every facilitator knows their group, their band, and their room.
- Kits counted and laid out per group before doors open.
- Registration desk has the sign-in sheet, badges, and allergy list.
- Timekeeper owns the master timetable and calls transitions out loud.
- Floating helpers know which groups they cover during breaks.
- Showcase space and closing certificates ready before the last block.

Organiser Notes

Projects were chosen to fit your notes where possible: *Focus the whole camp on Teachable Machine — each group trains their own image classifier and builds a simple project around it. Kids are beginners.*